

Why the 2-Selmer ranks agree

$y^2 = x^3 + a$ and $y^2 = x^3 - a^{-1}$ share their mod-2 Galois module; what that explains, and what it leaves to check

on a question of Rene Pannekoek, MathOverflow 227987 (2015)

1. The question

Numerically, the Mordell–Weil ranks of $y^2 = x^3 + p^3$ and $y^2 = x^3 - p^3$ almost always agree, the first exceptions being $p = 37, 61, 157, 193$. The curves are **not isogenous**, so the correlation is puzzling. Looking instead at 2-Selmer ranks makes the pattern exact: they appear to agree for **every** prime $p > 2$ — including at $p = 37$, where the ranks are 0 and 2 but both 2-Selmer ranks are 3. The general form conjectured is:

Conjecture. For a an **odd** integer, the elliptic curves

$$E_a : y^2 = x^3 + a \quad \text{and} \quad E_{-1/a} : y^2 = x^3 - a^{-1} \cong y^2 = x^3 - a^5$$

have equal 2-Selmer ranks.

The original statement is the case $a = p^3$; the variants with $x^3 \pm p$ against $x^3 \mp p^5$, and $x^3 \pm p^2$ against $x^3 \mp p^4$, are $a = \pm p$ and $a = \pm p^2$.

Note that $a \mapsto -1/a$ is an **involution**, and that it is the natural one: the curves $y^2 = x^3 + a$ are the sextic twists of one another, the twisting parameter lives in $\mathbb{Q}^\times/(\mathbb{Q}^\times)^6$, and $-1/a \equiv -a^5$ there.

2. The answer to “why are they correlated”

The 2-Selmer group of E is cut out of $H^1(\mathbb{Q}, E[2])$ by local conditions. So two curves with **isomorphic mod-2 Galois modules** have their 2-Selmer groups inside the **same** group, and their ranks are constrained to be close whether or not the curves are isogenous. That is exactly what happens here, and it is visible in one line.

Theorem A. Let $a \in \mathbb{Q}^\times$ and let $A_b = \mathbb{Q}[t]/(t^3 + b)$. Then

$$\theta \mapsto -1/\theta$$

is an isomorphism of $\mathbb{Q}$ -algebras $A_a \rightarrow A_{-1/a}$. Consequently

$$E_a[2] \cong E_{-1/a}[2]$$

as $G_{\mathbb{Q}}$ -modules, and the two 2-Selmer groups are subgroups of one and the same $H^1(\mathbb{Q}, M)$.

Proof. If $\theta^3 = -a$ then $(-1/\theta)^3 = -1/\theta^3 = 1/a$, so $-1/\theta$ is a root of $t^3 - 1/a$; the map is a $\mathbb{Q}$ -algebra map between 3-dimensional algebras sending a generator to a generator, hence an isomorphism. For an elliptic curve $y^2 = f(x)$ with f cubic, $E[2]$ is the $G_{\mathbb{Q}}$ -set of roots of f — the non-zero points are $(\theta_i, 0)$ — with its induced $\mathbb{F}_2$ -structure. An isomorphism of the étale algebras is an isomorphism of the root sets as $G_{\mathbb{Q}}$ -sets, hence of the modules. ■

So the curves, though non-isogenous, have the same 2-division field $\mathbb{Q}(\sqrt[3]{a}, \zeta_3)$ and the same mod-2 representation. Checked numerically on $a = 3, 5, 7, 9, 11, 13, 15, 25, 27, -5, -7, 49, 121$: the factorisation types of $x^3 + a$ and $x^3 - a^5$ agree at every good prime up to 300, and the 2-division fields are isomorphic.

This already answers the question as asked. “Why would the ranks of non-isogenous curves be correlated?” — because rank correlation is not what is happening. The **2-Selmer** ranks are correlated because the two curves share the object that 2-Selmer ranks are computed from. The Mordell–Weil

ranks then agree whenever $\text{III}[2]$ does not intervene, and $p = 37$ is precisely a case where it does: the ranks are 0 and 2, the 2-Selmer ranks both 3, the difference being $\dim \text{III}[2] = 2$ on one side and 0 on the other.

3. What remains: a purely local statement

Fix $M = E_a[2] = E_{-1/a}[2]$ and write $L_v(E) \subseteq H^1(\mathbb{Q}_v, M)$ for the image of $E(\mathbb{Q}_v)/2$ under the Kummer map δ_v . The first thing to record is that the **dimension** of L_v is not a property of the curve at all — it depends only on M and on v .

Lemma B'. For every place v ,

$$\dim_{\mathbb{F}_2} L_v = \begin{cases} \dim M^{G_v} & v \text{ finite} \\ v \nmid 2 \\ \dim M^{G_2} + 1 & v = 2 \\ \dim M^{G_\infty} - 1 & v = \infty. \end{cases}$$

Since the two curves have the same M by Theorem A, this already gives $\dim L_v(E_a) = \dim L_v(E_{-1/a})$ at every place. Here are two proofs, because the second explains **why** the answer is curve-independent.

3.0.1. First proof: the structure of $E(\mathbb{Q}_v)$

The Kummer sequence

$$0 \rightarrow E(\mathbb{Q}_v)/2E(\mathbb{Q}_v) \xrightarrow{\delta_v} H^1(\mathbb{Q}_v, M) \rightarrow H^1(\mathbb{Q}_v, E)[2] \rightarrow 0$$

makes δ_v **injective**, so $\dim L_v = \dim_{\mathbb{F}_2} E(\mathbb{Q}_v)/2E(\mathbb{Q}_v)$, and it is that group we must count.

For K a finite extension of $\mathbb{Q}_p$, the formal group gives a finite-index subgroup of $E(K)$ isomorphic to $(\mathfrak{m}_K, +) \cong \mathbb{Z}_p^{[K:\mathbb{Q}_p]}$, whence an isomorphism of topological groups

$$E(K) \cong \mathbb{Z}_p^{[K:\mathbb{Q}_p]} \times T, \quad T \text{ finite}.$$

Now divide by 2. The finite part contributes $\dim T/2T = \dim T[2] = \dim E(K)[2] = \dim M^{G_K}$, a finite abelian group having equal 2-rank in $T/2T$ and $T[2]$. The free part contributes $\dim \mathbb{Z}_p^{[K:\mathbb{Q}_p]}/2 = 0$ when p is **odd**, since 2 is then invertible in $\mathbb{Z}_p$, and $[K:\mathbb{Q}_p]$ when $p = 2$. For $K = \mathbb{Q}_v$ this is the first two lines.

For $v = \infty$: $E(\mathbb{R})$ is a compact real Lie group of dimension 1, so $E(\mathbb{R}) \cong \mathbb{R}/\mathbb{Z}$ or $\mathbb{R}/\mathbb{Z} \times \mathbb{Z}/2$ according as the real locus has one or two components. In the first case $E(\mathbb{R})$ is 2-divisible and $E(\mathbb{R})/2 = 0$, while $E(\mathbb{R})[2] = \mathbb{Z}/2$; in the second $E(\mathbb{R})/2 = \mathbb{Z}/2$ and $E(\mathbb{R})[2] = (\mathbb{Z}/2)^2$. Both times $\dim E(\mathbb{R})/2 = \dim M^{G_\infty} - 1$, the point being that a real cubic always has at least one real root, so $M^{G_\infty} \neq 0$.

3.0.2. Second proof: maximal isotropy and the Euler characteristic

This one is shorter and shows the shape of the answer. By Tate local duality, the Weil pairing makes M self-dual and the local condition L_v is **its own annihilator** under the induced pairing on $H^1(\mathbb{Q}_v, M)$ — the classical fact that the image of the Kummer map is maximal isotropic. Hence

$$\dim L_v = \frac{1}{2} \dim H^1(\mathbb{Q}_v, M).$$

For finite v the local Euler characteristic formula gives

$$h^0 - h^1 + h^2 = -[\mathbb{Q}_v : \mathbb{Q}_2] \dim M,$$

with the convention $[\mathbb{Q}_v : \mathbb{Q}_2] = 0$ for $v \nmid 2$; and local duality together with self-duality of M gives $h^2 = h^0$. So

$$h^1 = 2h^0 + [\mathbb{Q}_v : \mathbb{Q}_2] \dim M = 2 \dim M^{G_v} + 2[\mathbb{Q}_v : \mathbb{Q}_2],$$

using $\dim M = 2$, and halving returns $\dim L_v = \dim M^{G_v} + [\mathbb{Q}_v : \mathbb{Q}_2]$ — the first two lines at once. The archimedean place is not covered by that formula, and is the one place where the count has to be done by hand, as above.

Every quantity on the right-hand side — M^{G_v} , $[\mathbb{Q}_v : \mathbb{Q}_2]$, $\dim M$ — is an invariant of the **Galois module**, not of the curve. That is the structural reason two curves with the same M have local conditions of the same dimension at every place, and it is why the comparison in $(\star)$ is about which subspace, never about how big. A worked instance: at $v = 2$ we shall find $\dim M^{G_2} = 1$, so $\dim L_2 = 2$ and $|L_2| = 4$ — which is exactly what the computation of Section 7.3 returns, for both curves and all four classes of a .

Both $L_v(E_a)$ and $L_v(E_{-1/a})$ are therefore maximal isotropic subspaces of the same dimension in the same $H^1(\mathbb{Q}_v, M)$. So the conjecture would follow at once from

$$(\star) \quad L_v(E_a) = L_v(E_{-1/a}) \text{ for every place } v.$$

and $(\star)$ is automatic at all but finitely many places:

- **good v :** both conditions are the unramified subgroup $H_{\text{ur}}^1(\mathbb{Q}_v, M)$, which depends only on M ;
- **$v = \infty$:** $\Delta(y^2 = x^3 + k) = -432k^2 < 0$ for both curves, so both real loci are connected, $E(\mathbb{R})$ is divisible by 2, and $L_\infty = 0$ on both sides.

The bad places are the same for both curves — 2, 3 and the primes dividing a — so everything reduces to a finite check there.

4. The two descent maps differ by -1

There is one more piece of structure, and it says the two descents are not merely parallel but **aligned**.

Theorem B. Under the identification of Theorem A, the $\overline{\mathbb{Q}}$ -isomorphism between the two curves carries the descent map of E_a to λ times that of $E_{-1/a}$, where

$$\lambda = -1/\theta^2 = -(1/\theta)^2 \equiv -1 \quad \text{in } A_a^\times / (A_a^\times)^2.$$

Proof. $x^3 - a^{-1} = x^3 + au$ with $u = -a^{-2}$, so $E_{-1/a}$ is the sextic twist of E_a by u and the isomorphism over $\overline{\mathbb{Q}}$ is $(x, y) \mapsto (u^{1/3}x, u^{1/2}y)$. It carries roots to roots, so $\theta' = u^{1/3}\theta$, whence $u^{1/3} = \theta'/\theta = -1/\theta^2$; and then $X - \theta' = u^{1/3}(x - \theta)$. Finally $-1/\theta^2$ is -1 times a square. ■

Two readings. First, the quadratic part of the twist is by $u^{1/2} = a^{-1}\sqrt{-1}$: the involution $a \mapsto -1/a$ is **the quadratic twist by -1 , composed with a cubic twist** — and for $a = p^3$ the cubic part is trivial, so the original question is exactly about the quadratic twist by -1 . Second, χ_{-1} is ramified only at 2 and ∞ , which is why the two curves are so tightly coupled: at every odd place the twist is **unramified**.

5. Where the hypothesis “ a odd” lives

The conjecture is stated for odd a , and the computations say the restriction is not cosmetic. For odd a — positive and negative, prime, prime power and composite — the 2-Selmer ranks agree in every case tested. For even a they usually do **not**, and the failure is always by exactly one:

a	$\dim \text{Sel}_2(x^3 + a)$	$\dim \text{Sel}_2(x^3 - a^5)$	
2	1	0	differ by 1
6	0	1	differ by 1
8	2	1	differ by 1

a	$\dim \text{Sel}_2(x^3 + a)$	$\dim \text{Sel}_2(x^3 - a^5)$	
10	1	0	differ by 1
12	1	0	differ by 1
14	0	1	differ by 1
16	0	1	differ by 1
18	1	0	differ by 1

Fourteen of the sixteen even a tested disagree, every time by exactly 1. Since the local conditions agree at every good place and at ∞ regardless of parity, and since 2 is the only place where the quadratic part of the twist is ramified, this locates the whole content of the conjecture at $v = 2$: **oddness of a is what makes $L_2(E_a) = L_2(E_{-1/a})$** . A discrepancy of exactly 1 is what one expects from a single place where two maximal isotropic subspaces of a common $H^1(\mathbb{Q}_2, M)$ meet in a hyperplane.

6. Numerical verification

Sage's `selmer_rank` is $\dim_{\mathbb{F}_2} \text{Sel}_2$; in PARI, `ellrank` returns $[r_1, r_2, s, L]$ with $C = T + r_2 + s$, where $T = \dim E(\mathbb{Q})[2]$ is 1 when $x^3 + a$ has a rational root and 0 otherwise.

```
dimtors2(E) = { my(f = factor(x^3 + E.a6));
  sum(i = 1, #f~, if(poldegree(f[i,1]) == 1, 1, 0)); }
C2(k) = { my(E = ellinit([0,0,0,0,k]), r);
  r = ellrank(E);
  if(dimtors2(E) > 0, 1, 0) + r[2] + r[3]; }
```

family	agree / total	comment
odd a , $ a \leq 25$	26 / 26	reproduces the original table
odd a , mixed prime powers and composites	19 / 19	3, 5, ..., 81, 99, 121, 125
odd $a < 0$	12 / 12	-3, ..., -49
even a	2 / 16	always off by exactly 1

7. Completing the proof

Everything now rests on $(\star)$, and the engine is the following, which is proved outright.

Lemma C. Under the identification $\theta \mapsto \theta' = -1/\theta$ of Theorem A, the two descent maps **agree on 2-torsion**: for every i ,

$$\delta'(T'_i) = \delta(T_i) \quad \text{in } A^\times / (A^\times)^2.$$

Proof. On the i -th coordinate the value is the derivative: $g'(\theta'_i) = 3\theta_i^2$, against $f'(\theta_i) = 3\theta_i^2$, and their ratio is $(\theta'_i/\theta_i)^2$, a square. On the j -th coordinate, $j \neq i$,

$$\theta'_i - \theta'_j = (-1/\theta_i) + (1/\theta_j) = (\theta_i - \theta_j)/(\theta_i\theta_j),$$

and $\theta_i\theta_j = \theta_i^2\zeta$ for a cube root of unity ζ , because the three roots of $t^3 + a$ are θ_i times the cube roots of unity. Now θ_i^2 is a square and $\zeta = (\zeta^2)^2$ is a square, ζ having **odd** order 3. So the ratio is a square and the two coordinates agree. ■

That gives $(\star)$ wherever L_v is spanned by the image of the 2-torsion, and it settles three of the four ranges of places.

7.1. The archimedean place, and the good places

At a good v both conditions are $H_{\text{ur}}^1(\mathbb{Q}_v, M)$, which depends only on M . At $v = \infty$ there is nothing to check at all: with $G = \text{Gal}(\mathbb{C}/\mathbb{R})$ acting on $M = \{0, A, B, C\}$ by fixing A and swapping B, C , one has $M^G = \{0, A\}$, $(\sigma - 1)M = \{0, B + C\} = \{0, A\}$ and $\ker N = \{0, A\}$, so $H^1(\mathbb{R}, M) = 0$.

7.2. The place $v = 3$

Here $\dim L_3 = \dim E[2](\mathbb{Q}_3) \leq 1$, since two independent rational 2-torsion points would force $\zeta_3 \in \mathbb{Q}_3$, which is false. If the dimension is 0 both conditions vanish. If it is 1, then L_3 is the line spanned by $\delta(T)$ **provided** $\delta(T) \neq 1$ — and it is, because the $\mathbb{Q}_3$ -component of $\delta(T)$ is

$$f'(\theta_1) = 3\theta_1^2 \equiv 3,$$

and 3 is a uniformiser at 3, hence not a square. So $L_3 = \langle \delta(T) \rangle = \langle \delta'(T') \rangle = L'_3$ by Lemma C.

7.3. The place $v = 2$: a finite, exhaustive check

This is where the hypothesis “ a odd” lives, and where the argument becomes a computation — but a **complete** one, not a sample.

Over $\mathbb{Q}_2$ every unit is a cube, since cubing is bijective on $\mathbb{Z}_2^\times$. So the cubic part of any sextic twist by a unit is trivial, and for a odd the curve E_a over $\mathbb{Q}_2$ depends only on the class of a in $\mathbb{Q}_2^\times/(\mathbb{Q}_2^\times)^2$ — that is, only on $a \bmod 8$. **Four classes, and checking all four settles every odd a .**

`selmer-local2.gp` computes both conditions inside the single algebra $A_2 = \mathbb{Q}_2 \times K$, $K = \mathbb{Q}_2(\zeta)$ unramified quadratic, using the identification $\theta' = -1/\theta$: the descent value of E' at X is $X + 1/\theta$ in the $\mathbb{Q}_2$ -component and $(X - 1/\theta) - (1/\theta)\zeta$ in the K -component. Both come out as groups of order 4 — which is $\dim L_2 = \dim M^{G_2} + 1 = 2$, as it must be — and they **coincide**, for $a \equiv 1, 3, 5, 7$ modulo 8.

($\star$) holds at $v = 2$ for every odd a .

Two wrong turns, both caught by parity. Comparing $L_2(y^2 = x^3 + a)$ with $L_2(y^2 = x^3 - a)$ under the **naïve** identification — cube root against cube root — reports a disagreement, but it is meaningless: that identification differs from $\theta \mapsto -1/\theta$ by an automorphism of $E[2]$, so it compares subspaces of two different copies of H^1 . And a slip in the 2-torsion entry for E' , where $\theta'_1 - \theta'_2 = (\zeta^2 - 1)/\theta = (-2 - \zeta)/\theta$, inflated the group to order 8 — impossible, since $\dim L_2 = 2$. Both were exposed by the parity constraint: the comparison formula of Section 8 forces the Selmer ranks to differ by an odd number if 2 is the **only** place of discrepancy, and the ranks are equal.

7.4. The primes dividing a

The last range. Everything is decided by two valuations.

Proposition E. Let v be an odd prime, $v \neq 3$, with $v \mid a$. Then $L_v(E_a) = L_v(E_{-1/a})$.

Proof. Sixth powers change neither curve, so we may take $m = v(a)$ with $1 \leq m \leq 5$.

(i) If $3 \nmid m$, a root of $t^3 + a$ would have valuation $m/3 \notin \mathbb{Z}$, so $E[2](\mathbb{Q}_v) = 0$ and $\dim L_v = \dim E[2](\mathbb{Q}_v) = 0$. Both conditions vanish.

(ii) So $m = 3$; write $a = v^3 a_0$ with $v \nmid a_0$. A root lies in $\mathbb{Q}_v$ only if a_0 is a cube in $\mathbb{Z}_v^\times$; if it is not, again $L_v = 0 = L'_v$.

(iii) Otherwise let $\theta_1 \in \mathbb{Q}_v$ be that root, so $v(\theta_1) = 1$, the others being $\theta_1\zeta$ and $\theta_1\zeta^2$. Two valuations decide everything:

$$v(f'(\theta_i)) = v(3\theta_i^2) = 2 \quad \text{is EVEN} \quad (v \neq 3),$$

$$v(\theta_i - \theta_j) = v(\theta_1(\zeta^r - \zeta^s)) = 1 \quad \text{is ODD} \quad (i \neq j),$$

the second because $N(1 - \zeta) = (1 - \zeta)(1 - \zeta^2) = 3$ is a **unit** at v , so $\zeta^r - \zeta^s$ is a unit. Now split on v modulo 3.

If $v \equiv 2 \pmod{3}$ then $\zeta \notin \mathbb{Q}_v$, $t := \dim L_v = 1$, and $A_v = \mathbb{Q}_v \times F$ with $F = \mathbb{Q}_v(\zeta)$ unramified quadratic. The F -component of $\delta(T_1)$ is $\theta_1(1 - \zeta)$, of **odd** valuation, hence not a square; so $\delta(T_1) \neq 1$ and it spans the line L_v .

If $v \equiv 1 \pmod{3}$ then $\zeta \in \mathbb{Q}_v$, $t = 2$, and $A_v = \mathbb{Q}_v^3$. Reading off valuation parities coordinate by coordinate,

$$\delta(T_1) \mapsto (0, 1, 1), \quad \delta(T_2) \mapsto (1, 0, 1),$$

which are independent over $\mathbb{F}_2$; so $\delta(T_1), \delta(T_2)$ span the 2-dimensional L_v .

In both cases the image of the 2-torsion is all of L_v , and Lemma C identifies it with the image of the 2-torsion of E' , which is all of L'_v . Hence $L_v = L'_v$. ■

The same valuation argument is what settled $v = 3$ in Section 7.2, there in the simpler form that 3 is a uniformiser, so that $v_3(3\theta_1^2)$ is **odd** and $\delta(T)$ cannot be trivial. Note that the two cases of (iii) are exactly the fibres of type I_0^* with $c_v = 2$ and $c_v = 4$: the local image is the 2-torsion of the component group, and it is spanned by the 2-torsion of the curve — the situation Proposition 9 of the companion survey describes in general.

8. Status

Proved here. Theorem A — the two curves have isomorphic mod-2 Galois modules, by the explicit algebra isomorphism $\theta \mapsto -1/\theta$. This is what makes the correlation structural rather than accidental, and it is what the question was asking for. Theorem B — the descent maps differ by -1 modulo squares, and the involution is the quadratic twist by -1 composed with a cubic twist, so that at every **odd** place the twist is unramified. The reduction of the conjecture to the local statement $(\star)$, together with its verification at the good places and at ∞ .

Also proved (Section 7). Lemma C: the two descent maps agree on 2-torsion, by an explicit computation with $\theta_i\theta_j = \theta_i^2\zeta$ and the fact that ζ has odd order. With it, $(\star)$ holds at every good place, at ∞ (where $H^1(\mathbb{R}, M) = 0$), at $v = 3$ (where $\dim L_3 \leq 1$ and $\delta(T)$ has $\mathbb{Q}_3$ -component the class of the uniformiser 3, hence is non-trivial), and — by a finite and **exhaustive** computation, four classes covering every odd a — at $v = 2$.

Theorem. Let a be an odd integer. Then

$$\text{Sel}_2(y^2 = x^3 + a) = \text{Sel}_2(y^2 = x^3 - a^{-1})$$

as subgroups of $H^1(\mathbb{Q}, M)$; in particular the two 2-Selmer ranks are equal.

Proof. Theorem A identifies the two mod-2 modules, so both Selmer groups are cut out of the same $H^1(\mathbb{Q}, M)$ by local conditions. Those conditions agree at every place: at ∞ because $H^1(\mathbb{R}, M) = 0$; at every good place because both are H_{ur}^1 ; at $v = 3$ by Section 7.2; at the odd $v \mid a$ by Proposition E; and at $v = 2$ by the exhaustive computation of Section 7.3, four square classes covering every odd a . Equal local conditions cut out equal Selmer groups. ■

The one computational input is Section 7.3, and it is finite and exhaustive rather than a sample: over $\mathbb{Q}_2$ the curve depends only on a modulo 8. A hand proof there would make the argument entirely free of machine computation, and is the only thing still worth doing.

Where the hypothesis is used. Oddness of a enters exactly once, at $v = 2$: it is what makes the sextic twist parameter a **unit**, hence a cube, hence reduces $E_a/\mathbb{Q}_2$ to a quadratic twist class depending only on $a \bmod 8$. For even a that fails, and so does the conclusion — 14 of the 16 even a tested have Selmer ranks differing by exactly 1.

A remark on what “equal” costs. Comparing two Selmer structures on a self-dual module, the Greenberg–Wiles formula gives only

$$\dim \operatorname{Sel}(L) \equiv \dim \operatorname{Sel}(L') + \sum_v \dim(L_v/(L_v \cap L'_v)) \pmod{2}.$$

So parity agreement already requires the local discrepancies to cancel modulo 2, and the observed **equality** is the stronger statement that they vanish one by one. That is why $(\star)$, and not a parity argument, is the right target.